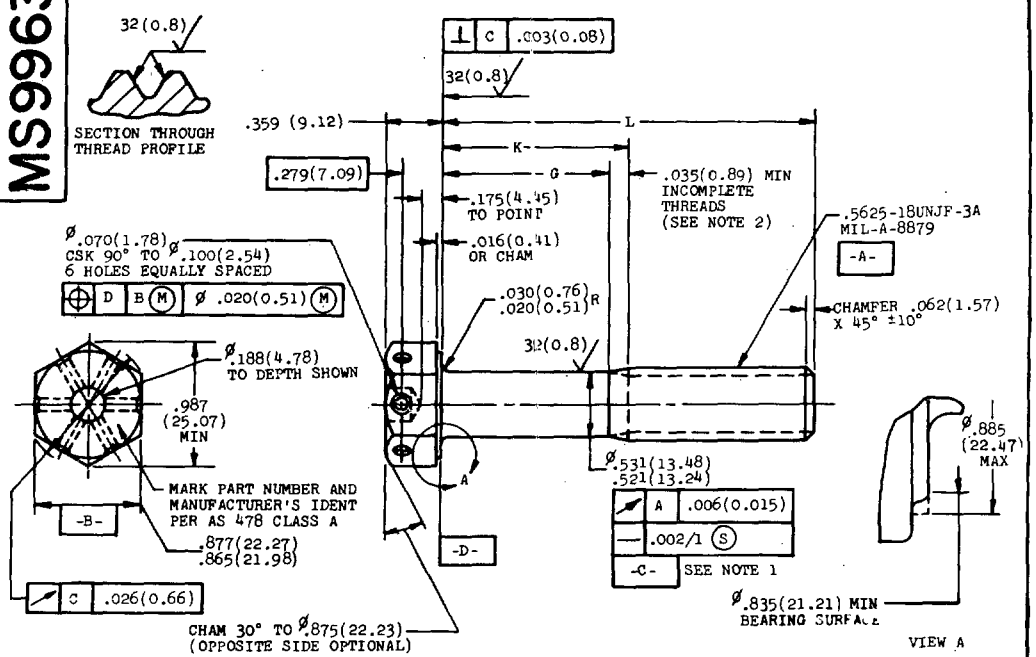
FED. SUP CLASS
5306

MS9963



1. FOR PART NUMBERS MS9963-04 THRU MS9963-25 THE THREAD PD SHALL REPLACE DATUM C.
2. INCOMPLETE THREADS NOT TO ENTER FILLET.
3. MATERIAL: STEEL AMS 6322.
4. HARDNESS: ROCKWELL C26-32.
5. FINISH: CADMIUM PLATE AMS 2400. DIMENSIONS SPECIFIED ARE AFTER PLATING.
6. MANUFACTURING SPECIFICATION: AMS 7452 EXCEPT HEAD SHALL BE UPSET.
7. HEAD TO SHANK FILLET SHALL BE COLD WORKED.
8. MAGNETIC PARTICLE INSPECTION PER AMS 2640 AFTER PLATING.
9. SURFACE TEXTURE: ANSI B46.1-1962. UNLESS OTHERWISE SPECIFIED, SURFACES TO BE 125 MICROINCHES (3.2 MICROMETERS) EXCEPT UPSET HEAD.
10. BREAK SHARP EDGES .003-.015 (0.08-0.38) UNLESS OTHERWISE SPECIFIED.
11. DIMENSIONS IN INCHES. METRIC CONVERSIONS ARE IN PARENTHESES. UNLESS OTHERWISE SPECIFIED: TOLERANCES; LINEAR DIMENSIONS ±.010 (0.25), ANGULAR DIMENSIONS ±5°. INTERNATIONAL SYSTEM UNITS (SI) SHOWN ARE FOR REFERENCE ONLY..
12. DIMENSIONING AND TOLERANCING: ANSI Y14.5-1966, Ø = DIAMETER.
13. DO NOT USE UNASSIGNED PART NUMBERS.

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P.A. Air Force - 11 Other Cust Army - AV Navy - AS	TITLE BOLT, MACHINE, HEXAGON HEAD, DRILLED, 6 HOLE, PD SHANK, STEEL AMS 6322, CADMIUM PLATED, .5625-18UNJF-3A	MILITARY STANDARD MS9963
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 1 OF 2

DD FORM 672-1
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PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

5306-0498

APPROVED 27 Jun 74 REVISED

User activities: Army, Air Force, DSA

Review activities: Army, Navy, Air Force, DSA

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MS9963

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PART NUMBER	L IN.	L (mm)	G MIN IN.	G MIN (mm)	K MAX IN.	K MAX (mm)	APPROX MASS LB/100	APPROX MASS kg/100
MS9963-04	.875	22.22	.030	0.77	.134	3.40	10.61	4.813
MS9963-05	.938	23.82	.030	0.77	.134	3.40	11.01	4.994
MS9963-06	1.000	25.40	.030	0.77	.134	3.40	12.41	5.629
MS9963-07	1.062	26.97	.030	0.77	.134	3.40	12.81	5.810
MS9963-08	1.125	28.58	.030	0.77	.134	3.40	13.21	5.992
MS9963-09	1.188	30.18	.030	0.77	.134	3.40	13.60	6.169
MS9963-10	1.250	31.75	.030	0.77	.134	3.40	13.98	6.341
MS9963-11	1.312	33.32	.030	0.77	.134	3.40	14.36	6.514
MS9963-12	1.375	34.92	.030	0.77	.134	3.40	14.75	6.690
MS9963-13	1.438	36.52	.030	0.77	.134	3.40	15.13	6.863
MS9963-14	1.500	38.10	.030	0.77	.134	3.40	15.51	7.035
MS9963-15	1.562	39.67	.030	0.77	.188	4.77	15.89	7.208
MS9963-16	1.625	41.28	.079	2.01	.250	6.35	16.28	7.384
MS9963-17	1.688	42.88	.141	3.59	.312	7.92	16.66	7.557
MS9963-18	1.750	44.45	.204	5.19	.375	9.52	17.04	7.729
MS9963-19	1.812	46.02	.267	6.79	.438	11.12	17.43	7.906
MS9963-20	1.875	47.62	.329	8.36	.500	12.70	17.81	8.078
MS9963-21	1.938	49.22	.392	9.96	.562	14.27	18.19	8.251
MS9963-22	2.000	50.80	.454	11.54	.625	15.88	18.58	8.428
MS9963-23	2.125	53.98	.579	14.71	.750	19.05	19.34	8.772
MS9963-24	2.250	57.15	.704	17.89	.875	22.22	20.11	9.122
MS9963-25	2.375	60.32	.829	21.06	1.000	25.40	20.87	9.466
MS9963-26	2.500	63.50	.954	24.24	1.125	28.57	21.64	9.816
MS9963-27	2.625	66.68	1.079	27.41	1.250	31.75	22.41	10.165
MS9963-28	2.750	69.85	1.204	30.59	1.375	34.92	23.17	10.510
MS9963-29	2.875	73.02	1.329	33.76	1.500	38.10	23.94	10.859
MS9963-30	3.000	76.20	1.454	36.94	1.625	41.27	24.71	11.208

PART NUMBER	L IN.	L (mm)	G MIN IN.	G MIN (mm)	K MAX IN.	K MAX (mm)	APPROX MASS LB/100	APPROX MASS kg/100
MS9963-31	3.125	79.38	1.579	40.11	1.750	44.45	25.47	11.553
MS9963-32	3.250	82.55	1.704	43.29	1.875	47.62	26.24	11.902
MS9963-33	3.375	85.72	1.829	46.46	2.000	50.80	27.00	12.247
MS9963-34	3.500	88.90	1.954	49.64	2.125	53.97	27.77	12.596
MS9963-35	3.625	92.08	2.079	52.81	2.250	57.15	28.54	12.946
MS9963-36	3.750	95.25	2.204	55.99	2.375	60.32	29.30	13.290
MS9963-37	3.875	98.42	2.329	59.16	2.500	63.50	30.07	13.940
MS9963-38	4.000	101.60	2.454	62.34	2.625	66.67	30.84	13.989
MS9963-39	4.125	104.78	2.579	65.51	2.750	69.85	31.60	14.334
MS9963-40	4.250	107.95	2.704	68.69	2.875	73.02	32.37	14.683
MS9963-41	4.375	111.12	2.829	71.86	3.000	76.20	33.13	15.028
MS9963-42	4.500	114.30	2.954	75.04	3.125	79.37	33.90	15.377
MS9963-43	4.625	117.48	3.079	78.21	3.250	82.55	34.67	15.726
MS9963-44	4.750	120.65	3.204	81.39	3.375	85.72	35.43	16.071
MS9963-45	4.875	123.82	3.329	84.56	3.500	88.90	36.20	16.420
MS9963-46	5.000	127.00	3.454	87.74	3.625	92.07	36.96	16.765
MS9963-47	5.125	130.18	3.579	90.91	3.750	95.25	37.73	17.114
MS9963-48	5.250	133.35	3.704	94.09	3.875	98.42	38.50	17.463
MS9963-49	5.375	136.52	3.829	97.26	4.000	101.60	39.26	17.808
MS9963-50	5.500	139.70	3.954	100.44	4.125	104.77	40.03	18.157
MS9963-51	5.625	142.88	4.079	103.61	4.250	107.95	40.80	18.507
MS9963-52	5.750	146.05	4.204	106.79	4.375	111.12	41.56	18.851
MS9963-53	5.875	149.22	4.329	109.96	4.500	114.30	42.33	19.200
MS9963-54	6.000	152.40	4.454	113.14	4.625	117.47	43.09	19.545

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FED. SUP CLASS 5306

APPROVED 27 Jun 74 REVISED

P.A. Air Force - 11
Other Cust
Army - AV
Navy - AS

TITLE
BOLT, MACHINE, HEXAGON HEAD, DRILLED, 6 HOLE, PD SHANK, STEEL AMS 6322, CADMIUM PLATED, .5625-18UNJF-3A

MILITARY STANDARD
MS9963

PROCUREMENT SPECIFICATION

SUPERSEDES:

SHEET 2 OF 2

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